



Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

Re-evaluation of rate constants for the reaction $\text{N}_2\text{H}_4 (+ \text{M}) \rightleftharpoons \text{NH}_2 + \text{NH}_2 (+ \text{M})$

Carlos J. Cobos^a, Peter Glarborg^b, Paul Marshall^c, Jürgen Troe^{d,*}

^a INIFTA, Facultad de Ciencias Exactas, Universidad Nacional de La Plata, CONICET, La Plata, Argentina

^b Department of Chemical and Biochemical Engineering, Technical University of Denmark, Denmark

^c Department of Chemistry and Center for Advanced Scientific Computing and Modeling, University of North Texas, Denton, TX, USA

^d Max-Planck-Institut für Multidisziplinäre Naturwissenschaften, and Institut für Physikalische Chemie, Universität Göttingen, Tammannstrasse 6, D 37077 Göttingen, Germany

ARTICLE INFO

Article history:

Received 1 April 2022

Revised 29 August 2022

Accepted 11 September 2022

Available online xxx

Keywords:

Nitrogen chemistry in combustion

ABSTRACT

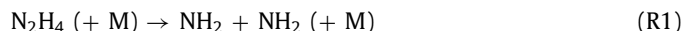
Rate constants for the dissociation/recombination reaction $\text{N}_2\text{H}_4 (+ \text{M}) \rightleftharpoons \text{NH}_2 + \text{NH}_2 (+ \text{M})$ are determined by a combination of quantum-chemical calculations and statistical unimolecular rate theory. Between 1100 and 2500 K, limiting low-pressure rate constants for hydrazine dissociation in the bath gas Ar of $k_0 = [\text{Ar}] 6.1 \times 10^{20} (T/1000 \text{ K})^{-7.3} \exp(-34490 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$, limiting high-pressure rate constants of $k_\infty = 7.6 \times 10^{16} (T/1000 \text{ K})^{-1.0} \exp(-33600 \text{ K}/T) \text{ s}^{-1}$, and center broadening factors of the falloff curves (between 1100 and 1600 K in Ar) of $F_{\text{cent}} = 0.71 \exp(-T/1460 \text{ K}) + 0.29 \exp(-T/21 \text{ K}) + \exp(-13400 \text{ K}/T)$ were calculated. Using equilibrium constants $K_c = 1.7 \times 10^3 (T/1000 \text{ K})^{-1.5} \exp(-33,460 \text{ K}/T) \text{ mol cm}^{-3}$, between 300 and 600 K limiting low-pressure rate constants for the reverse recombination of NH_2 radicals in the bath gas N_2 of $k_{\text{rec},0} = [\text{N}_2] 4.4 \times 10^{20} (T/400 \text{ K})^{-6.9} \exp(-1630 \text{ K}/T) \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1}$, limiting high-pressure rate constants of $k_{\text{rec},\infty} = 4.4 \times 10^{13} (T/1000 \text{ K})^{0.44} \exp(-140 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$, and center broadening factors (between 300 and 600 K in N_2) of $F_{\text{cent}} = \exp(-T/1130 \text{ K}) + \exp(-10340 \text{ K}/T)$ were obtained. A comparison with experimental results for hydrazine dissociation from the literature suggests incomplete falloff extrapolations toward k_∞ and experimental problems in the determination of k_0 at temperatures above about 1600 K. Implications of the present re-evaluated rate constants for the modeling of high temperature ammonia oxidation kinetics are discussed, showing an only small influence of their precise values on the overall properties of the process.

© 2022 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Nitrogen chemistry in combustion has been extensively investigated over the last 50 years due mainly to its implications for pollutant emissions. Amine reactions play a key role in the conversion of reactive nitrogen species at high temperature. Different from hydrocarbons, oxidation of amines involves a selectivity of reactions in that they may form nitrogen oxides or reduce nitrogen oxides to molecular nitrogen. Much of our present knowledge of amine combustion chemistry can be attributed to the work of Miller and coworkers [1–8]. In this field, reactions of amidogen radicals (NH_2) play a central role. It appears, therefore, important to regularly review such reactions.

A convenient experimental source of amidogen radicals has been the thermal dissociation of hydrazine



which is the issue of the present article. Hydrazine is also used as a long-term storable propellant, as well as a precursor to polymerization catalysts, pharmaceuticals, and agrochemicals and, therefore, has been studied extensively. The ignition and combustion properties of N_2H_4 , e. g., have been discussed in [9], as well as its propensity for vapor detonation in [10]. The chemistry of hydrazine has recently received renewed attention due to the emergence of ammonia as a carbon-free energy carrier [11–14]. When ammonia is used as a fuel in engines or gas turbines, owing to the combination of high concentrations and elevated pressures, considerable amounts of hydrazine may form through NH_2 radical recombination,



Hydrazine dissociation R1 has been studied over a wide range of pressures and temperatures in shock tubes [15–22] and flow reactors [23–25]. The data were evaluated by Konnov and De Ruyck

* Corresponding author.

E-mail address: juergen.troe@mpinat.mpg.de (J. Troe).

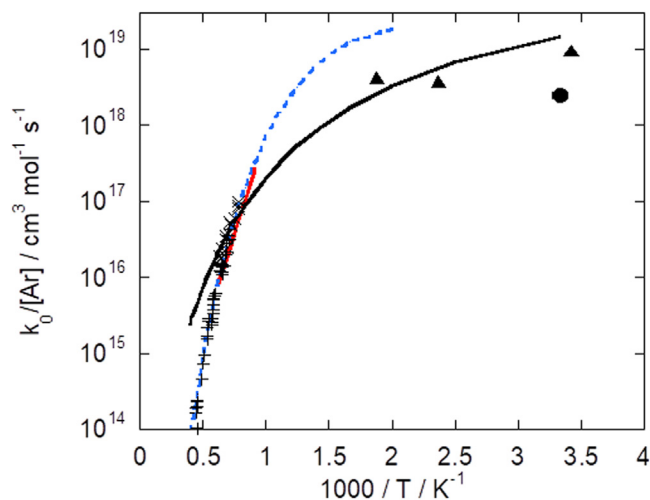


Fig. 1. Data for limiting low-pressure third-order rate constants $k_{\text{rec},0}/[\text{Ar}]$ of the reaction $\text{NH}_2 + \text{NH}_2 (+ \text{Ar}) \rightarrow \text{N}_2\text{H}_4 (+ \text{Ar})$. The symbols denote the experimental data of Khe et al. [26] (●) and Altinay and Macdonald [33] (▲) for the forward reaction and data derived from shock wave studies of the reverse reaction by Diesen [17] (+) and Meyer et al. [21] (×) converted through the equilibrium constant (the results from [33], obtained for $M = \text{N}_2$, were corrected for the difference in collision efficiencies [34]). The blue and red lines denote fits to the experimental data by Konnov and De Ruyck [9] (solid red), and Dai et al. [36] (blue dashes), respectively. The theoretical determination by Klippenstein et al. [35] is given in solid black.

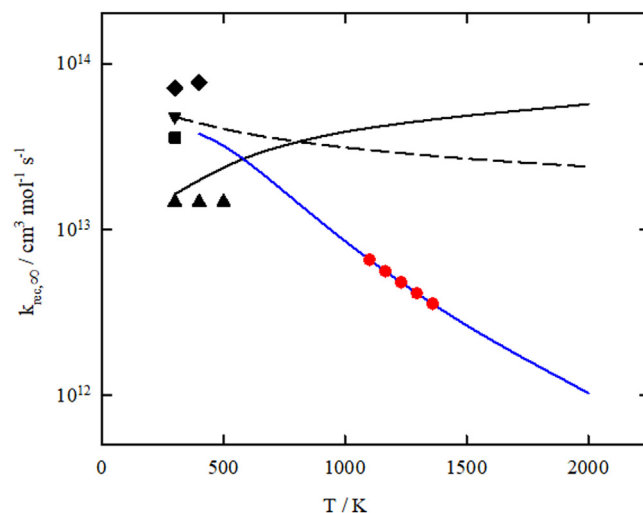


Fig. 2. Data for limiting high-pressure second-order rate constants $k_{\text{rec},\infty}$ of the reaction $\text{NH}_2 + \text{NH}_2 \rightarrow \text{N}_2\text{H}_4$. The symbols denote experimental data of Khe et al. [26] (●), Lozovskii et al. [27] (■), Sarkisov et al. [28] (▼), and Fagerström et al. [29] (◆) for the forward reaction and data derived from shock wave studies of the reverse reaction (red dots) by Meyer et al. [21] and Genich et al. [22] converted through the equilibrium constant by Dai et al. [36]. The blue line represents the results recommended by Dai et al. [36]. The dashed black line is from the theoretical results by Klippenstein et al. [35] and the solid black line from the SACM/CT modeling of the present work.

[9], who proposed a set of pressure-dependent rate constants for the reaction. The rate constant k_2 for the reverse R2 of reaction R1 has been investigated at low temperature by several groups [26–33]. The results show significant scatter, which can be attributed in part to uncertainty in the rate constant for the side reaction $\text{NH}_2 + \text{H} (+ \text{M}) \rightarrow \text{NH}_3 (+ \text{M})$ [32,33]. Reactions R1 and R2 were also studied theoretically by Glarborg et al. [34]. The various pathways of reaction R2 have, furthermore, been analyzed by Klippenstein et al. [35], while third body efficiencies of various collision partners M relative to N_2 were predicted in [34].

Apart from the role of secondary reactions, the pressure dependence of reactions R1 and R2 presents the problem of derivation of their full falloff curves, which would allow for an extrapolation to proper limiting low- and high-pressure rate constants. For instance, Fig. 1 compares reported results for limiting low-pressure third-order rate constants of reaction R2, including low-temperature data of Khe et al. [26] and Altinay and Macdonald [33], high temperature measurements of the rate constant of the reverse reaction R1 by Diesen [17] and Meyer et al. [21], converted through the equilibrium constant, and theoretical values by Klippenstein et al. [35]. Furthermore, fits to experimental results by Konnov and De Ruyck [9] and Dai et al. [36] are shown. While the theoretical rate constants from Klippenstein et al. [35] agree well with the experimental data at low temperatures and also with results for R1 at temperatures around 1500 K, they seem to markedly overpredict the experimental data at higher temperatures. On the other hand, the rate constants from Konnov and De Ruyck [9] and Dai et al. [36] comply with the shock tube data, but extrapolate poorly to low temperatures. The discrepancy between the theoretical rate constants for $\text{NH}_2 + \text{NH}_2 (+ \text{M}) \rightleftharpoons \text{N}_2\text{H}_4 (+ \text{M})$ and the high temperature shock tube results for the reverse reaction cannot be explained by uncertainties in the thermodynamic data. Recent evaluations indicate that the properties of both NH_2 [8] and N_2H_4 [37,38] are known with high accuracy.

Similar inconsistencies between low- and high-temperature rate constants for reactions R1 and R2, as illustrated for the low-pressure limit in Fig. 1, are also observed for the high-pressure limit. This is shown in Fig. 2 which compares limiting high-

pressure rate constants for reaction R2 from Khe et al. [26], Lozovskii et al. [27], Sarkisov et al. [28], and Fagerström et al. [29] at low temperatures with high-temperature data derived by Dai et al. [36] from hydrazine dissociation experiments of Meyer et al. [21] and Genich et al. [22]. Theoretical results by Klippenstein et al. [35] are also shown together with theoretical results from the present work.

The differences between experimental and theoretical results in Figs. 1 and 2 call for a re-evaluation, both of the experimental data and of the theoretical approach. In view of the progress made since the early work, the original shock wave experiments of hydrazine dissociation have been critically re-examined. The theoretical determination of the rate constants has also been repeated, using more recent molecular parameters for NH_2 and N_2H_4 and employing state-of-the-art unimolecular rate theory. This leads to an improved representation of the temperature- and pressure-dependence of the rate constants. Wherever differences to the experimental data are encountered, we try to trace their origin.

2. Theoretical determination of rate constants

The present analysis follows the procedure previously described, e. g., for the reactions $\text{C}_2\text{F}_4 (+ \text{M}) \rightleftharpoons \text{CF}_2 + \text{CF}_2 (+ \text{M})$ in [39] or $\text{CF}_3 (+ \text{M}) \rightleftharpoons \text{CF}_2 + \text{F} (+ \text{M})$ in [40]. It first determines the limiting low-pressure rate constants k_0 by the method of [41–43]. The limiting high-pressure rate constants k_∞ then are obtained by combination of the Statistical Adiabatic Channel Model with Classical Trajectory (SACM/CT) calculations as proposed in [44]. As shown later, the latter method leads to results for k_∞ similar to those from Variable Reaction Coordinate/Transition State Theory (VRC/TST) as used in [35]. The transition of the rate constants k from k_0 to k_∞ , i. e. the falloff curve of the reaction, finally is represented in reduced form k/k_∞ as a function of k_0/k_∞ with the approximate expressions from [45,46] which were derived from solutions of appropriate master equations. The modeling of both k_0 and k_∞ requires a quantum-chemical characterization of the Minimum Energy Path (MEP) potential as well as of the changes of molecular parameters along the MEP. The modeling of k_0 , in addition,

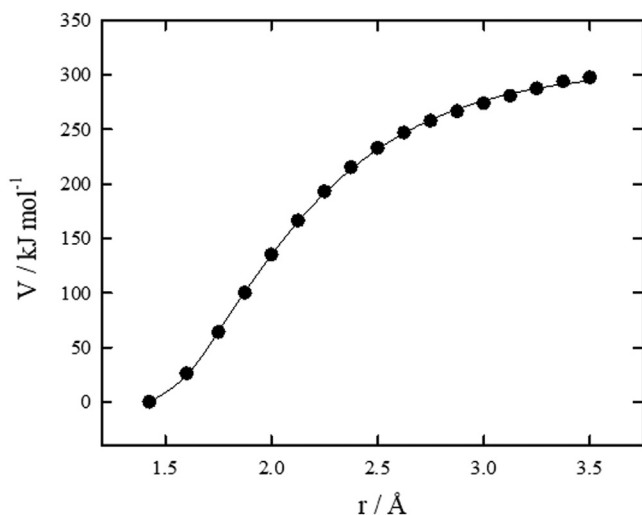


Fig. 3. Electronic potential for $\text{N}_2\text{H}_4 \rightarrow \text{NH}_2 + \text{NH}_2$ along the MEP, derived at the G4//M06-2X/6-311++G(3d2f,3pd) level of theory and fitted to a Morse function, see text.

requires knowledge on collisional energy transfer such as the average energy $\langle \Delta E \rangle$ transferred per collision, to be obtained either by fitting to the experiments, by theoretical calculations such as described in [34], or by estimation using a “standard value” [39,40].

2.1. Quantum-chemical calculations

To obtain the MEP potential, the variation of energy with the N-N bond length in hydrazine was investigated computationally. First, optimized molecular structures and harmonic vibrational frequencies were derived at the M06-2X/6-311++G(3d2f,3pd) level of density functional theory (DFT) [47]. The calculations were performed using the Gaussian 09 suite of programs with the default integration grid [48]. Then, single-point energies were obtained at these structures with G4 composite ab initio theory [49] and fitted to a Morse function as shown in Fig. 3. The Morse parameters are $r_e = 1.425 \text{ Å}$, $D_e = 306.6 \text{ kJ mol}^{-1}$, and $\beta = 1.90 \text{ Å}^{-1}$. Inclusion of vibrational zero-point energy yields $D_0 = 267.4 \text{ kJ mol}^{-1}$, in excellent accord with the literature values given in Section 2.6.

As the N-N bond breaks, a number of vibrational modes disappear and the corresponding transitional frequencies along the MEP are plotted in Fig. 4. It may be seen that their variation with the N-N distance r is nearly proportional to $\exp(-\alpha r)$, where α may be termed the “looseness parameter” [43].

Rotational constants for the reactants and products (as well as vibrational frequencies) are also given in Section 2.6. The variation of the average of the two smallest rotational constants of hydrazine as the N-N bond lengthens was obtained from the DFT calculations noted above, and is plotted in Fig. 5. The results were fitted by the function $(B + C)/2 = 0.819 \text{ cm}^{-1} / [1 + 0.9336 (r - 1.425 \text{ Å}) + 0.4376 (r - 1.425 \text{ Å})^2]$ with r in Å.

2.2. Calculation of limiting high-pressure rate constants $k_{\text{rec},\infty}$ and k_∞

Limiting high-pressure rate constants for the recombination $\text{NH}_2 + \text{NH}_2$ ($k_{\text{rec},\infty}$) and for N_2H_4 dissociation (k_∞) were evaluated following the SACM/CT method of [44]. In outline, phase-space theory yields an upper limit to the limiting high-pressure recombination rate constant, which then is modified by a rigidity factor (smaller than unity) to account for anisotropy in the inter-fragment potential, i. e.

$$k_{\text{rec},\infty} = f_{\text{rigid}} k_{\text{rec},\infty}^{\text{PST}} \quad (1)$$

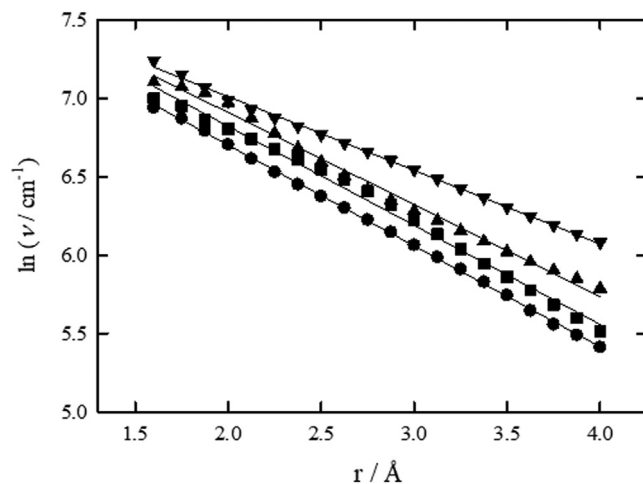


Fig. 4. Transitional frequencies along the MEP, calculated at the M06-2X/6-311++G(3d2f,3pd) level of theory. The fitted curves correspond to decays proportional to $\exp(-\alpha r)$ with looseness parameters $\alpha = 0.64$ (•), 0.62 (■), 0.59 (▲) and 0.47 (▼) Å^{-1} .

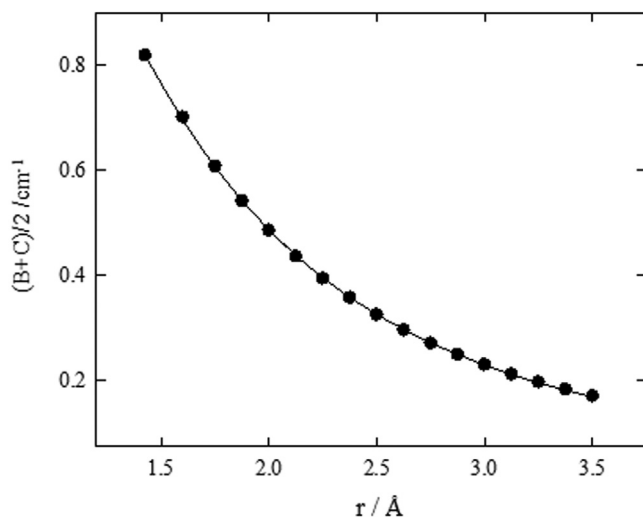


Fig. 5. Rotational constants for N_2H_4 , calculated along the MEP with M06-2X/6-311++G(3d2f,3pd) theory. The expression for the fitted curve is given in the text.

The rigidity factor f_{rigid} is based on the parameters α , β , and on the rotational properties of the fragments and the adduct. The dissociation rate coefficient is obtained as $k_\infty = k_{\text{rec},\infty} \times K_C$ where K_C is the dissociation equilibrium constant. The results are listed in Table 1. Approximate analytical representations of the results are

$$K_C = 1.7 \times 10^3 (T/1000 \text{ K})^{-1.5} \exp(-33460 \text{ K}/T) \text{ mol cm}^{-3} \quad (2)$$

$$k_{\text{rec},\infty} = 4.4 \times 10^{13} (T/1000 \text{ K})^{0.44} \exp(-140 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1} \quad (3)$$

$$k_\infty = 7.6 \times 10^{16} (T/1000 \text{ K})^{-1.0} \exp(-33600 \text{ K}/T) \text{ s}^{-1} \quad (4)$$

2.3. Calculation of limiting low-pressure rate constants $k_{\text{rec},0}$ and k_0

Limiting low-pressure rate constants for dissociation are expressed in the form given in [43]

$$k_0 = [M] \beta_c Z_{\text{LJ}} (\rho_{\text{vib},h}(E_0) RT/Q_{\text{vib}}) \exp(-E_0/RT) F_{\text{anh}} F_E F_{\text{rotint}} F_{\text{rot}} \quad (5)$$

Table 1

Limiting high-pressure rate constants for the reactions $\text{NH}_2 + \text{NH}_2 \rightarrow \text{N}_2\text{H}_4$ and $\text{N}_2\text{H}_4 \rightarrow \text{NH}_2 + \text{NH}_2$, $k_{\text{rec},\infty}$ and k_{∞} , respectively, obtained by following the SACM/CT method of [44] as well as equilibrium constants $K_{\text{C}} = k_{\infty} / k_{\text{rec},\infty}$, calculated with the molecular parameters given in Section 2.6, see text.

T/K	$k_{\text{rec},\infty}^{\text{PST}}$ $\text{cm}^3\text{mol}^{-1}\text{s}^{-1}$	f_{rigid}	$k_{\text{rec},\infty}$ $\text{cm}^3\text{mol}^{-1}\text{s}^{-1}$	K_{C} mol cm^{-3}	k_{∞} s^{-1}
300	1.2×10^{14}	0.14	1.6×10^{13}	3.7×10^{-45}	6.1×10^{-32}
600	1.4×10^{14}	0.19	2.7×10^{13}	2.0×10^{-21}	5.5×10^{-8}
1000	1.6×10^{14}	0.24	3.9×10^{13}	5.4×10^{-12}	2.1×10^2
1500	1.7×10^{14}	0.28	4.8×10^{13}	2.1×10^{-7}	1.0×10^7
2000	1.9×10^{14}	0.31	5.7×10^{13}	3.5×10^{-5}	2.0×10^9
2500	1.9×10^{14}	0.32	6.1×10^{13}	6.6×10^{-4}	4.0×10^{10}

Table 2

Limiting low-pressure rate constants for $\text{NH}_2 + \text{NH}_2 (+ \text{Ar}) \rightarrow \text{N}_2\text{H}_4 (+ \text{Ar})$ and $\text{N}_2\text{H}_4 (+ \text{Ar}) \rightarrow \text{NH}_2 + \text{NH}_2 (+ \text{Ar})$, $k_{\text{rec},0} / [\text{Ar}]$ and $k_0 / [\text{Ar}]$, respectively, see text.

T/K	$k_{\text{rec},0}^{\text{SC}} / [\text{Ar}]$ $\text{cm}^6\text{mol}^{-2}\text{s}^{-1}$	$k_0^{\text{SC}} / [\text{Ar}]$ $\text{cm}^3\text{mol}^{-1}\text{s}^{-1}$	β_{c}	$k_{\text{rec},0} / [\text{Ar}]$ $\text{cm}^6\text{mol}^{-2}\text{s}^{-1}$	$k_0 / [\text{Ar}]$ $\text{cm}^3\text{mol}^{-1}\text{s}^{-1}$
300	6.0×10^{19}	2.3×10^{-25}	0.23	1.4×10^{19}	5.2×10^{-26}
600	8.4×10^{18}	1.7×10^{-3}	0.13	1.1×10^{18}	2.2×10^{-4}
1000	1.7×10^{18}	9.0×10^6	0.080	1.3×10^{17}	7.2×10^5
1500	3.8×10^{17}	8.0×10^{10}	0.050	1.9×10^{16}	4.0×10^9
2000	1.1×10^{17}	4.0×10^{12}	0.033	3.8×10^{15}	1.3×10^{11}
2500	4.2×10^{16}	2.8×10^{13}	0.023	9.6×10^{14}	6.3×10^{11}

where the dissociation threshold energy E_0 , the Lennard-Jones collision frequency Z_{LJ} with the bath gas M, the harmonic vibrational density of states $\rho_{\text{vib},h}$ and the vibrational partition function Q_{vib} are derived with the data given in Section 2.6. The factors F_{anh} , F_{E} , F_{rotint} , and F_{rot} are calculated as described in [43]. In particular, F_{rot} reflects the influence of angular momentum (quantum number J) and is based on the evaluation of rotational barriers $E_0(J)$ on the calculated electronic potential and with the rotational constants of Section 2.1. The $E_0(J)$ were fitted to the form

$$E_0(J) - E_0(J=0) = C_{\text{v}}[J(J+1)]^{\nu} \quad (6)$$

For hydrazine dissociation, we derived $C_{\text{v}}/hc = 9.59 \times 10^{-3} \text{ cm}^{-1}$ and $\nu = 1.26$.

The strong collision rate constants k_0^{SC} correspond to a collision efficiency $\beta_{\text{c}} = 1$. β_{c} in our work is estimated with a typical value for the average (total) energy $\langle \Delta E \rangle_{\text{total}}$ transferred per collision of the excited molecules with the bath gas M. As concluded from earlier experiments over broad temperature ranges (see [39,40] and references therein) here again it is assumed to be given by $\langle \Delta E \rangle_{\text{total}}/hc = -100 \text{ cm}^{-1}$ and not to depend on the temperature. β_{c} is approximately related to $\langle \Delta E \rangle_{\text{total}}$ by [41,43]

$$\beta_{\text{c}} / (1 - \beta_{\text{c}}^{1/2}) = - \langle \Delta E \rangle_{\text{total}} / (F_{\text{E}}RT) \quad (7)$$

One should note that $\langle \Delta E \rangle_{\text{total}}$ is also related to the average energy $\langle \Delta E \rangle_{\text{down}}$ transferred per down-transition by [41]

$$\langle \Delta E \rangle_{\text{total}} = - \langle \Delta E \rangle_{\text{down}}^2 / (\langle \Delta E \rangle_{\text{down}} + F_{\text{E}}RT) \quad (8)$$

(one should also note that a temperature independent $\langle \Delta E \rangle_{\text{total}}$ according to Eq. (8) corresponds to a temperature dependent $\langle \Delta E \rangle_{\text{down}}$). One finally has the dissociation rate coefficient $k_0 = \beta_{\text{c}} k_0^{\text{SC}}$, while the recombination rate coefficient is found by conversion with the equilibrium constant $K_{\text{C}} = k_{\infty}/k_{\text{rec},\infty}$ from Eq. (2). The results are given in Table 2 for the bath gas M = Ar. It has to be emphasized that the value for $\langle \Delta E \rangle_{\text{total}}$ is a fit parameter, chosen to be of “standard magnitude”. In the “weak collision limit”, $\beta_{\text{c}} \ll 1$, k_0 is proportional to β_{c} ; uncertainties of $\langle \Delta E \rangle_{\text{total}}$ directly correspond to uncertainties of k_0 . A decrease of $\langle \Delta E \rangle_{\text{total}}$ and a change of its temperature dependence, in principle, thus would allow for a fit to Diesen’s high temperature results. However, the derived properties of $\langle \Delta E \rangle_{\text{total}}$ would appear highly unusual.

Approximate analytical representations of the results, for the bath gas M = Ar, are

$$k_{\text{rec},0} = [\text{Ar}] 6.8 \times 10^{17} (T/1000 \text{ K})^{-6.9} \exp(-1630 \text{ K}/T) \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1} \quad (9)$$

$$k_0 = [\text{Ar}] 6.1 \times 10^{20} (T/1000 \text{ K})^{-7.3} \exp(-34490 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1} \quad (10)$$

For the bath gas M = N₂, we obtained

$$k_{\text{rec},0} = [\text{N}_2] 4.4 \times 10^{20} (T/400 \text{ K})^{-6.9} \exp(-1630 \text{ K}/T) \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1} \quad (11)$$

$$k_0 = [\text{N}_2] 2.9 \times 10^{24} (T/400 \text{ K})^{-8.4} \exp(-35100 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1} \quad (12)$$

2.4. Calculation of rate constants at intermediate pressures

The rate constants k were interpolated between their limiting low- and high-pressure values as a function of the bath gas density $[\text{M}]$ using the falloff expression

$$k = F \times k_0 k_{\infty} / (k_{\infty} + k_0) \quad (13)$$

where k_0 is proportional to $[\text{M}]$, see Eqs. 9–(12). The “broadening factor” F as a function of the ratio $k_0/k_{\infty} = x$ is determined such as proposed in [43,45,46]. Its magnitude is dominated by center broadening factors F_{cent} , calculated following [43,50,51] to be

$$F_{\text{cent}} = 0.71 \exp(-T/1460 \text{ K}) + 0.29 \exp(-T/21 \text{ K}) + \exp(-13400 \text{ K}/T) \quad (14)$$

for the bath gas Ar between 1100 and 1600 K, and

$$F_{\text{cent}} = \exp(-T/1130 \text{ K}) + \exp(-10340 \text{ K}/T) \quad (15)$$

for the bath gas N₂ between 300 and 600 K. The detailed shape of F was fitted according to the recommendation of [45,46], i. e. in the form

$$F = (1 + x) / [1 + x^n]^{1/n} \quad (16)$$

Table 3

Parameters for falloff curves of the reaction $\text{N}_2\text{H}_4 (+ \text{Ar}) \rightarrow \text{NH}_2 + \text{NH}_2 (+ \text{Ar})$. A value of $\langle \Delta E \rangle_{\text{total}}/hc = -100 \text{ cm}^{-1}$ was used for the calculation of β_c .

T / K	$k_{\infty} / \text{s}^{-1}$	$k_0 / [\text{Ar}] \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$	β_c	F_{cent}
1110	4.9×10^3	1.0×10^7	0.071	0.34
1200	4.4×10^4	6.0×10^7	0.065	0.31
1300	3.5×10^5	3.0×10^8	0.059	0.29
1400	2.0×10^6	1.2×10^9	0.054	0.26
1550	1.9×10^7	5.9×10^9	0.047	0.25

Table 4

Parameters for falloff curves of the reaction $\text{NH}_2 + \text{NH}_2 (+ \text{N}_2) \rightarrow \text{N}_2\text{H}_4 (+ \text{N}_2)$. A value of $\langle \Delta E \rangle_{\text{total}}/hc = -100 \text{ cm}^{-1}$ was used for the calculation of β_c .

T / K	$k_{\text{rec}, \infty} / \text{s}^{-1}$	$k_{\text{rec}, 0} / [\text{N}_2] \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$	β_c	F_{cent}
300	1.6×10^{13}	1.4×10^{19}	0.23	0.77
422	2.2×10^{13}	6.4×10^{18}	0.18	0.69
533	2.6×10^{13}	2.8×10^{18}	0.15	0.62

where $n = [\ln 2 / \ln(2/F_{\text{cent}})][0.8 + 0.2x^q]$, $q = (F_{\text{cent}} - 1) / \ln(F_{\text{cent}}/10)$. Alternatively, broadening factors could be approximated by the relationship

$$\log F = \log F_{\text{cent}} / \{1 + (\log x/N)^2\} \quad (17)$$

where $N = 0.75 - -1.27 \log F_{\text{cent}}$ [43].

All falloff curves, including those based either on the limiting rate constants of [35] or those modeled in Sections 2.2 and 2.3, were calculated using Eq. (16). Examples of F_{cent} are included in Tables 3 and 4 (for conditions of the described dissociation and recombination experiments).

2.5. Recommended rate constants

The calculated results for the limiting high-pressure rate constants of $\text{NH}_2 + \text{NH}_2$ recombination and N_2H_4 dissociation, respectively, may be summarized as

$$k_{\text{rec}, \infty} = 4.4 \times 10^{13} (T/1000 \text{ K})^{0.44} \exp(-140 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$$

$$k_{\infty} = 7.6 \times 10^{16} (T/1000 \text{ K})^{-1.0} \exp(-33600 \text{ K}/T) \text{ s}^{-1}$$

For the bath gas Ar, our recommended limiting low-pressure rate constants are

$$k_{\text{rec}, 0} = [\text{Ar}] 6.8 \times 10^{17} (T/1000 \text{ K})^{-6.9} \exp(-1630 \text{ K}/T) \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1}$$

$$k_0 = [\text{Ar}] 6.1 \times 10^{20} (T/1000 \text{ K})^{-7.3} \exp(-34490 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$$

For the bath gas N_2 , the corresponding expressions are

$$k_{\text{rec}, 0} = [\text{N}_2] 4.4 \times 10^{20} (T/400 \text{ K})^{-6.9} \exp(-1630 \text{ K}/T) \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1}$$

$$k_0 = [\text{N}_2] 2.9 \times 10^{24} (T/400 \text{ K})^{-8.4} \exp(-35100 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$$

Rate constants in the falloff range are obtained by using expressions $k = F \times k_0 k_{\infty} / (k_{\infty} + k_0)$ with $x = k_0 / k_{\infty}$ and broadening factors F , given by $F = (1 + x)/[1 + x^n]^{1/n}$ where $n = [\ln 2 / \ln(2/F_{\text{cent}})][0.8 + 0.2x^q]$, $q = (F_{\text{cent}} - 1) / \ln(F_{\text{cent}}/10)$ from [45,46] (or alternatives like Eq. (17) from [43]).

Center broadening factors F_{cent} are given by

$$F_{\text{cent}} = 0.71 \exp(-T/1460 \text{ K}) + 0.29 \exp(-T/21 \text{ K}) + \exp(-13400 \text{ K}/T)$$

for the bath gas Ar between 1100 and 1600 K, or

$$F_{\text{cent}} = \exp(-T/1130 \text{ K}) + \exp(-10340 \text{ K}/T)$$

for the bath gas N_2 between 300 and 600 K.

2.6. Molecular parameters used in the modeling

Bond dissociation enthalpy for $\text{N}_2\text{H}_4 \rightarrow \text{NH}_2 + \text{NH}_2$ at 0 K: $\Delta H_{\text{f},0}^{\circ} (E_0) = 266.6 \text{ kJ mol}^{-1}$ (from the enthalpies of formation $\Delta H_{\text{f},0}^{\circ}(\text{N}_2\text{H}_4) = 111.57 \pm 0.47 \text{ kJ mol}^{-1}$ and $\Delta H_{\text{f},0}^{\circ}(\text{NH}_2) = 189.1 \pm 1.0 \text{ kJ mol}^{-1}$, see [57]).

N_2H_4 vibrational frequencies (cm^{-1}): 377, 3332, 3189, 1806, 1283, 1098, 871, 3332, 3332, 1608, 1324 and 1042 [58]. The torsional mode 377 cm^{-1} was treated as a hindered rotor [42] with a rotational barrier of 1100 cm^{-1} [59].

N_2H_4 rotational constants (cm^{-1}): 4.803, 0.8015 and 0.8010 [60].

NH_2 vibrational frequencies (cm^{-1}): 1497 [54], 3219 and 3301 [61,62].

NH_2 rotational constants (cm^{-1}): 23.70, 12.95 and 8.17 [63].

Lennard-Jones parameters: $\sigma(\text{N}_2\text{H}_4) = 3.62 \text{ \AA}$ and $\varepsilon/k(\text{N}_2\text{H}_4) = 138 \text{ K}$ [64]; $\sigma(\text{Ar}) = 3.47 \text{ \AA}$, $\varepsilon/k(\text{Ar}) = 114 \text{ K}$ [65]; $\sigma(\text{N}_2) = 3.74 \text{ \AA}$, $\varepsilon/k(\text{N}_2) = 82 \text{ K}$ [65].

3. Comparison of experimental and theoretical rate constants

Fig. 2 includes the theoretical high-pressure rate constants for reaction R2 from the SACM/CT calculations of the present work and from VRC-TST of Klippenstein et al. [35] (represented by $k_{\text{rec}, \infty} = 5.62 \times 10^{14} T^{-0.414} \exp(-33 \text{ K}/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$). The two theoretical results apparently agree within experimental accuracy. One should note that reaction R2, under low-temperature conditions, is mostly investigated close to the high-pressure limit of the reaction and only limited falloff extrapolations toward $k_{\text{rec}, \infty}$ had to be made. This is different from high-temperature studies of the dissociation reaction R1 where considerably higher bath gas pressures have to be applied to allow for a satisfactory extrapolation toward k_{∞} . The difference between the modeled k_{∞} and the experimental values then may be attributed to incomplete falloff extrapolation in the shock wave experiments. This is illustrated in Fig. 6 where experimental falloff curves from dissociation experiments are compared with theoretical falloff curves as derived in

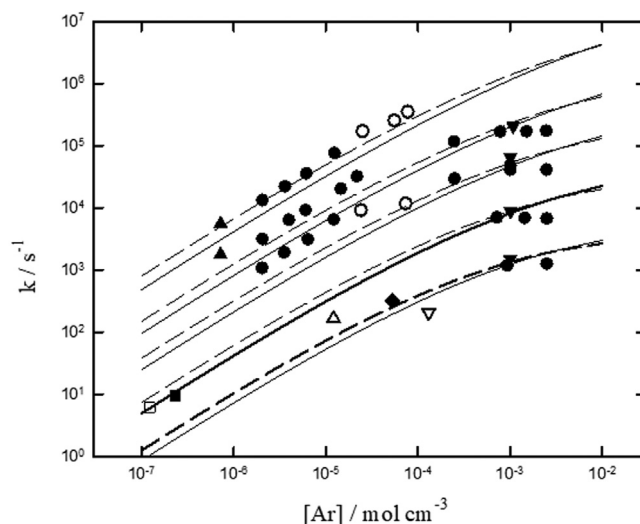


Fig. 6. Falloff curves for $\text{N}_2\text{H}_4 (+ \text{Ar}) \rightarrow \text{NH}_2 + \text{NH}_2 (+ \text{Ar})$. Representations from bottom to top, for 1110, 1200, 1300, 1400 and 1550 K. Experimental data: Szwarc [23] ($\square$, with $M = \text{toluene}$, converted to $M = \text{Ar}$ with collision efficiencies from [34]); Moberly [15] (∇); Diesen [17] ($\blacktriangle$); Kerr et al. [24] ($\blacksquare$, with $M = \text{toluene}$, converted to $M = \text{Ar}$ with collision efficiencies from [34]); Eberstein and Glassman [25] ($\triangle$, with $M = \text{N}_2$, converted to $M = \text{Ar}$ with collision efficiencies from [34]); Michel and Wagner [20] ($\circ$); McHale et al. [18] ($\blacklozenge$); Meyer et al. [21] ($\circ$); Genich et al. [22] ($\blacktriangledown$). Theoretical curves: dashed, —: Klippenstein et al. [35]; full, —: present work.

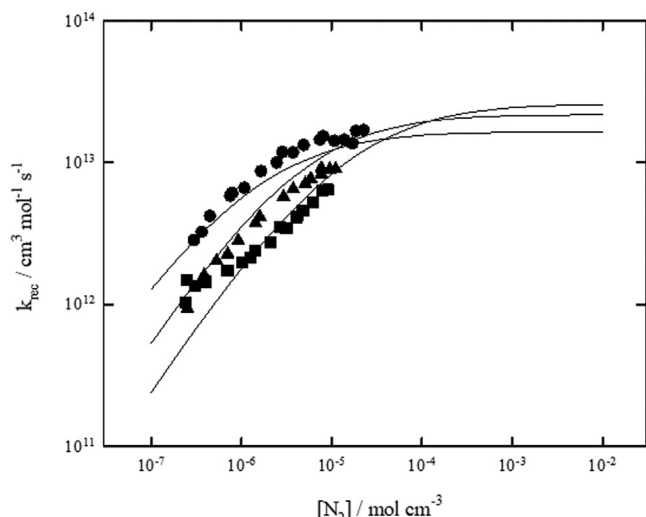


Fig. 7. Falloff curves for $\text{NH}_2 + \text{NH}_2 (+ \text{N}_2) \rightarrow \text{N}_2\text{H}_4 (+ \text{N}_2)$ at 292 K ($\circ$), 422 K ($\blacktriangle$) and 533 K ($\blacksquare$): experimental data from Altinay and Macdonald [33]. $-$: theoretical results from the present work with $\langle \Delta E \rangle_{\text{total}}/hc = -100 \text{ cm}^{-1}$.

Section 2.4. The figure shows that, even at the highest bath gas concentrations applied in [21,22], the reaction was still considerably below the high-pressure limit. The revised falloff extrapolations from the present approach according to Fig. 6, however, appear fully consistent with the values of k_∞ from eq. (4) as well as the theoretical results from Klippenstein et al. [35].

Falloff curves for reaction R2 are also represented well by the present calculations. This is illustrated in Fig. 7 for experiments between 292 and 533 K from Altinay and Macdonald [33]. Apparently, the choice of the standard value of $\langle \Delta E \rangle_{\text{total}}/hc = -100 \text{ cm}^{-1}$ (used both for $\text{M} = \text{Ar}$ and N_2) in the present calculations leads to realistic values of the rate constants.

There is a more serious problem with the limiting low-pressure rate constants k_0 for dissociation from Diesen [17] at temperatures of 2000 K and above. Fig. 8 illustrates that the calculated values

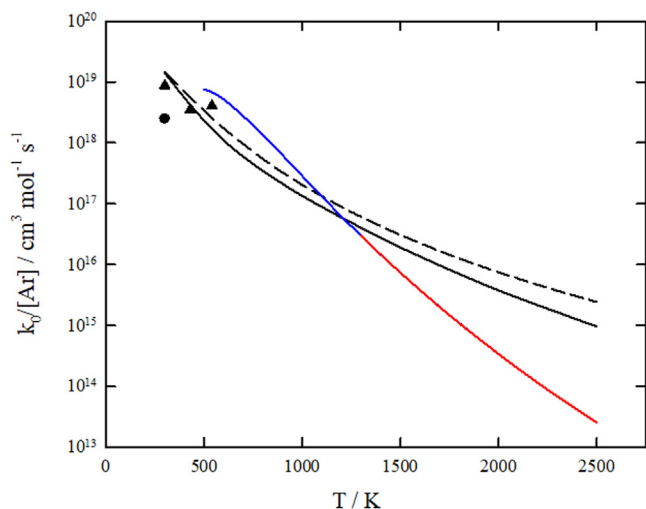


Fig. 8. Limiting low-pressure rate constants k_0 for $\text{N}_2\text{H}_4 (+ \text{Ar}) \rightarrow \text{NH}_2 + \text{NH}_2$ ($+ \text{Ar}$). Red curve: experimental data from Diesen [17], in agreement with experimental data from Meyer et al. [21] between 1200 and 1600 K, see Fig. 6. Experimental points from Khe et al. [26] ($\circ$) and Altinay and Macdonald [33] ($\blacktriangle$) converted through the equilibrium constant and represented by the blue/red curve by Dai et al. [36]. Theoretical calculations by Klippenstein et al. [35] with $\langle \Delta E \rangle_{\text{down}}/hc = 150 (T/300 \text{ K})^{0.85} \text{ cm}^{-1}$ (dashed black line) and from the present work with $\langle \Delta E \rangle_{\text{total}}/hc = -100 \text{ cm}^{-1}$ (solid black line).

from the present work as well as those from Klippenstein et al. [35] here are systematically above Diesen's data. As emphasized before, some uncertainty of using a standard value for $\langle \Delta E \rangle_{\text{total}}$ independent of temperature in the present treatment could not be avoided. However, the calculations of $\langle \Delta E \rangle_{\text{down}}$ in [34], such as used in [35], result in an even larger discrepancy between experimental and theoretical values of k_0 . We are, therefore, led to the conclusion that there was a problem in the experiments. In contrast to the experiments of [21], which used optical detection of the decomposing N_2H_4 , dissociation at low pressures and high temperatures behind the shock waves in [17] was followed by extracting gas through a pinhole, using time-of-flight (TOF) mass spectrometry for quantification. For the slowest rates (between about 1200 and 1500 K), the results by Diesen [17] practically coincided with those from [21], but for faster rates at higher temperatures the measured dissociation rates of hydrazine apparently were too slow. We attribute this to short-comings of the experimental technique used. At the time of these experiments, the probing of reaction progress behind shock waves through a pinhole and with TOF mass spectrometry was still in its infancy. While it was most attractive for analyzing the composition of the reacting mixtures, due to boundary layer effects in the probing pinhole it had limited accuracy for the time dependences of fast processes. Results obtained by Diesen for other dissociations like those of Cl_2 [52], F_2 [53], and NF_2 [54] with this technique showed similar deviations in comparison to experiments using other analysis techniques [55]. The difficulty in tracking fast reactions has been a systematic problem with TOF sampling through a pinhole and it was not solved until more recently [56]. For this reason, experimental data from [17] at temperatures above about 1600 K were discarded in the present work. Instead, the modeling results (assuming $\langle \Delta E \rangle_{\text{total}}$ to be temperature independent, see [66]) here were preferred.

4. Secondary reactions

The experimental studies of hydrazine dissociation in static and flow reactors as well as in shock waves were influenced by secondary reactions to a varying extent. On the one hand, these reactions have to be accounted for when conclusions on the rate constant k of reaction R1 are drawn. On the other hand, the observations enable one to test kinetic models of high temperature amine chemistry, relevant for contexts such as hydrazine pyrolysis and ammonia ignition.

The effect of secondary reactions on the derivation of k in shock wave experiments was partly reduced by increasingly diluting the reactant/bath gas mixtures. Nevertheless, a kinetic modeling is required to appreciate the role of secondary reactions. We base this analysis on the model reviewed in [8], with modifications from recent work on NH_3 pyrolysis [67,68] updated with results from the present work. Table 5 describes the kinetic model, as far as relevant to the pyrolysis of hydrazine.

As indicated in Table 5, the primary dissociation of hydrazine R1 may be followed by the fastest reaction with its dissociation product amidogen in the reaction



which is known to be fast [70]. If there were no other reactions, after an induction time this would lead to a tripling of the hydrazine decay rate. The corresponding induction times were documented, e. g., in Fig. 1 of [16,20]. At longer residence times, the increased consumption rate of hydrazine can be explained in terms of subsequent reactions of N_2H_3 leading to chain branching. Even though recent theoretical work [69–72] has improved our understanding of the N_2H_3 reaction subset, significant uncertainties remain.

Table 5

Selected reactions for hydrazine pyrolysis (rate constants $k = A T^n \exp(-E_a/RT)$ with T in K, E_a/R in K, A in $\text{K}^{-n} \text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$ for k , in $\text{K}^{-n} \text{s}^{-1}$ for k_∞ , and in $\text{K}^{-n} \text{mol cm}^{-3} \text{s}^{-1}$ for $k_0/[Ar]$). Results for R1 from present work (pw), for R4 from [69]; $F_{\text{cent}} = 0.71 \exp(-T/1460 \text{ K}) + 0.29 \exp(-T/21 \text{ K}) + \exp(-13400 \text{ K}/T)$ for R1 and $0.17 \exp(-T/80000 \text{ K}) + 0.84 \exp(-T/28 \text{ K}) + \exp(-7300 \text{ K}/T)$ for R4).

	Reaction	A	n	E_a/R	Ref.
R1	$\text{N}_2\text{H}_4 (+\text{Ar}) \rightarrow \text{NH}_2 + \text{NH}_2 (+\text{Ar})$ $k_{1,\infty}$	7.6×10^{19}	-1.000	33600	pw
	$k_{1,0}/[Ar]$	6.1×10^{42}	-7.330	34490	
R3	$\text{N}_2\text{H}_4 + \text{NH}_2 \rightarrow \text{N}_2\text{H}_3 + \text{NH}_3$	2.0×10^{01}	3.620	-397	[70]
R4	$\text{N}_2\text{H}_3 (+\text{Ar}) \rightarrow \text{HNNH} + \text{H} (+\text{Ar})$ $k_{4,\infty}$	1.3×10^{11}	0.819	24187	[69]
	$k_{4,0}/[Ar]$	3.8×10^{40}	-6.880	27407	
R5	$\text{N}_2\text{H}_4 + \text{H} \rightarrow \text{N}_2\text{H}_3 + \text{H}_2$	2.7×10^{05}	2.560	1218	[71]
R6	$\text{N}_2\text{H}_3 + \text{H} \rightarrow \text{NH}_2 + \text{NH}_2$	1.0×10^{14}	0.000	0	[72]
R7	$\text{N}_2\text{H}_3 + \text{NH}_2 \rightarrow \text{H}_2\text{NN} + \text{NH}_3$	1.1×10^{01}	3.080	106	[69]
R8	$\text{tHNNH} + \text{NH}_2 \rightarrow \text{NNH} + \text{NH}_3$	2.7×10^{05}	2.226	-520	[69]

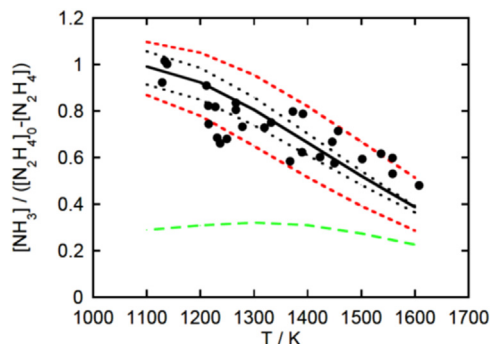
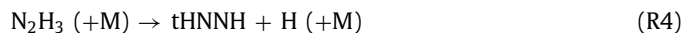


Fig. 9. Comparison of experimental and modeled NH_3 yields in the pyrolysis of hydrazine. Experimental data (symbols) from Michel and Wagner [20] were obtained from measurements behind incident and reflected shock waves. The yields were modeled for full conversion of hydrazine with the rate constants from Table 5 (black solid line), with a factor of 2 variation of k for hydrazine dissociation R1 (black dotted lines), with the reaction $\text{N}_2\text{H}_3 + \text{H} \rightarrow \text{NH}_2 + \text{NH}_2$ (R6) omitted (green long-dashed line), and with a factor of 3 variation of k_4 for $\text{N}_2\text{H}_3 (+\text{M})$ (R4, red short-dashed lines); see the text for the experimental conditions.

In the following, the sensitivity of experimental observations to some of the reactions shown in Table 5 will be illustrated. The most reliable results on the rate constant k of reaction R1 apparently were obtained in shock tube experiments at temperatures of 1100–1400 K. In this range, the choice of k among the values from Konnov and De Ruyck [9], Klippenstein et al. [35], and the present work, does not have a large impact on the modeling predictions, because the values of k anyway were all fitted to the experimental decay rates of hydrazine. More direct insight into secondary reactions, e. g., can be obtained from measurements of ammonia yields in hydrazine pyrolysis, such as described in [17,20] (the formation of other products like N_2 and H_2 was also monitored in [17]). A comparison of experimental ammonia yields from Michel and Wagner [20] with modeling results (using the rate constants from Table 5) is shown in Fig. 9. The figure summarizes results from experiments between 1100 and 1600 K, hydrazine concentrations in the range $[\text{N}_2\text{H}_4]_t = 0/[Ar] = (0.085\text{--}0.44)\%$, and bath gas concentrations in the range $[Ar] = (0.7\text{--}7.7) \times 10^{-5} \text{ mol cm}^{-3}$. The solid black line in the figure corresponds to modeling results from the present work. It satisfactorily agrees with the experimental data. The dependence on the value of k is relatively weak (the dotted black lines in the figure correspond to modeling results with k varied by a factor of 2). On the other hand, the dissociation of N_2H_3 (reaction R4) and the subsequent reaction of N_2H_4 with H atoms (reaction R5) have a stronger influence on the ammonia yield.

Since the experimental results in Fig. 9 were obtained under a wide range of reactant concentrations and pressures and since the fraction of hydrazine decomposed at the point of measurement was not stated, the data must be interpreted cautiously. Still, the results apparently validate the rate constants of key reactions in the N_2H_3 subset.

The analysis of modeling predictions indicates that, after the initial period of hydrazine pyrolysis, the unimolecular dissociation R1 is only a minor consumption channel for hydrazine; the consumption of N_2H_4 then happens largely by its reaction with NH_2 (R3) and H (R5), forming N_2H_3 . The major consumption steps for N_2H_3 are its thermal dissociation,



and the reactions of N_2H_3 with the radical pool, primarily atomic hydrogen. According to the theoretical work of [72], the reaction



proceeds without a barrier and is expected to be fast, while H-abstraction to produce an isomer of diazene and molecular hydrogen is only a minor channel. The ammonia yield in hydrazine pyrolysis then is controlled by the competition between reactions R4 and R6. Dissociation of N_2H_3 (R4) feeds into the diazene subset, yielding ultimately N_2 , while $\text{N}_2\text{H}_3 + \text{H}$ (R6) forms NH_2 radicals, subsequently leading to NH_3 by H-abstraction from N_2H_4 or diazene. As the temperature increases, the dissociation R4 becomes dominant compared to the reaction with H (R6) and the ammonia yield decreases. This is observed both in the experimental results and in the modeling. The results in Fig. 9 provide an experimental validation of the conclusion that the formation of amidogen is the dominant product channel for the reaction $\text{N}_2\text{H}_3 + \text{H}$ (R6) and that it is fast. A model without this step shows a very low ammonia yield, almost independent of temperature and in contrast to the experiments. The modeling also suggests that the rate coefficient of reaction R4 from the theoretical study of [69], under the conditions of Fig. 9, should be accurate to about a factor of 3.

The observation of NH_3 as a pyrolysis product has brought up again the question of the primary dissociation pathway of hydrazine [73]. A primary formation of ammonia by a reaction



would be energetically more favorable (by about 58 kJ mol^{-1}) than reaction R1. Meanwhile, the quantum-chemical calculations of [35] have shown that reaction R9 involves a substantial energy barrier in addition to the reaction enthalpy, leading to a higher threshold energy (of about 289 kJ mol^{-1}) than that of reaction R1 (about 267 kJ mol^{-1}) and that it proceeds over a rigid activated complex. An experimental test for reaction R9 in [73] was proposed by adding NO to dissociating hydrazine. On the one hand, NO is known to react with NH_2 to form $\text{N}_2 + \text{H}_2\text{O}$; on the other hand, NO may also react with NH to form $\text{N}_2\text{O} + \text{H}$. The detection of N_2O in [76] then could have been taken as evidence for the pathway R9. However, as NH can also be formed in the mechanism of secondary reactions following reaction R1, the observation of N_2O is no proof for a contribution of pathway R9 to hydrazine pyrolysis.

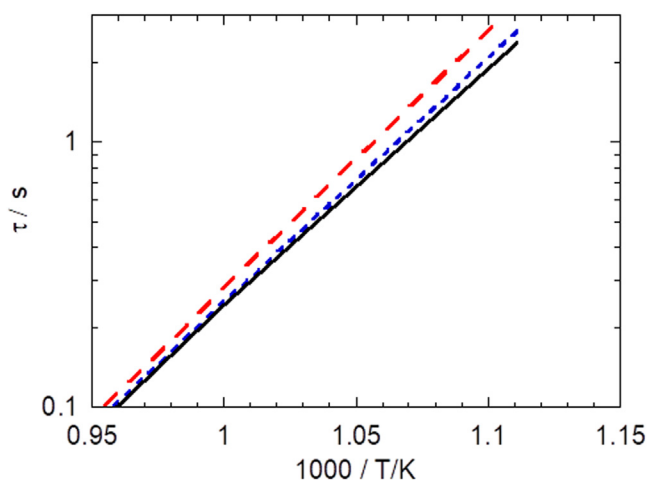


Fig. 10. Influence of the rate constant k for $\text{N}_2\text{H}_4 (+\text{M}) \rightarrow \text{NH}_2 + \text{NH}_2 (+\text{M})$ on predicted ignition delay times τ of a stoichiometric NH_3/air mixture at 100 atm (red long-dashed line: k from Konnov and De Ruyck [9], blue short-dashed line: k from Klippenstein et al. [35], black solid line: k from the present work).

One may finally ask for the relevance of reaction R1 to other reaction systems involving NH_3 . When ammonia is used as a fuel in engines or gas turbines, owing to the combination of high concentrations and elevated pressures, considerable amounts of hydrazine may form through NH_2 radical recombination,



Because reaction R2 is terminating, it may also serve to delay ignition of NH_3 , particularly at medium temperatures and elevated pressure. Fig. 10 shows the effect of the rate constant k of hydrazine dissociation (R1) on predictions of ignition delays of stoichiometric NH_3/air mixtures at 100 atm and temperatures of 900–1100 K, i. e., for conditions representative of a large 2-stroke marine diesel engine. The predictions with the rate constants for R1 from [35] and from the present work agree closely under these conditions, while using the value of Konnov and De Ruyck [9] results in a calculated ignition delay time, which is about 30% higher at 900 K. This difference is expected, since at 900 K the value from [9] is extrapolated outside its designated temperature range.

5. Conclusions

The described theoretical calculation of the rate constants for reactions R1 and R2 provides satisfactory agreement with the experimental data between 300 and about 1600 K. Discrepancies between experiments from [17] and theoretical results for higher temperatures were attributed to problems with the detection method used for fast kinetics, while problems with the derived limiting high-pressure rate constants from [21] were explained by incomplete falloff extrapolations. The consistency of experimental and theoretical rate constants again illustrates the benefit of using unimolecular rate theory in combination with quantum-chemical calculations of molecular parameters. The rate constants summarized in Section 2.5, therefore, are recommended for modeling nitrogen chemistry in combustion, although the influence of the rate constants of reactions R1 and R2 on the overall properties of the reaction system is only small.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This article is dedicated to the memory of James A. Miller who has contributed to the understanding of numerous reaction systems, including the present one. CC and JT acknowledge support by the Deutsche Forschungsgemeinschaft (Projekt TR 69/21-1 and TR 69/21-3). PG acknowledges funding from the Innovation Fund Denmark for the AEngine Grand Solutions project. PM was supported by the U.S. Department of Energy, Office of Basic Energy Sciences, Division of Chemical Sciences, Geosciences and Biosciences/Gas Phase Chemical Physics under Contract No. DESC0020952.69/

References

- [1] J.A. Miller, M.C. Branch, R.J. Kee, A chemical kinetic model for the selective reduction of nitric oxide by ammonia, *Combust. Flame* 43 (1981) 81–98.
- [2] J.A. Miller, M.D. Smooke, R.M. Green, R.J. Kee, Kinetic modeling of the oxidation of ammonia in flames, *Combust. Sci. Technol.* 34 (1983) 149–176.
- [3] J.A. Miller, C.T. Bowman, Mechanism and modeling of nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 15 (1989) 287–338.
- [4] P. Glarborg, K. Dam-Johansen, J.A. Miller, R.J. Kee, M.E. Coltrin, Modeling the thermal DeNOx process in flow reactors. Surface effects and nitrous oxide formation, *Int. J. Chem. Kinet.* 26 (1994) 421–436.
- [5] J.A. Miller, P. Glarborg, Modeling the thermal DeNOx process: closing in on a final solution, *Int. J. Chem. Kinet.* 31 (1999) 757–765.
- [6] J.A. Miller, M.J. Pilling, J. Troe, Unravelling combustion mechanisms through a quantitative understanding of elementary reactions, *Proc. Combust. Inst.* 30 (2005) 43–88.
- [7] S.J. Klippenstein, L.B. Harding, P. Glarborg, J.A. Miller, The role of NNH in NO formation and control, *Combust. Flame* 158 (2011) 774–789.
- [8] P. Glarborg, J.A. Miller, B. Ruscic, S.J. Klippenstein, Modeling nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 67 (2018) 31–68.
- [9] A.A. Konnov, J. De Ruyck, Kinetic modeling of the decomposition and flames of hydrazine, *Combust. Flame* 124 (2001) 106–126.
- [10] L. Catoire, J. Luche, G. Dupré, C. Paillard, Critical reactions for the hydrazine vapor detonations, *Shock Waves* 11 (2001) 97–103.
- [11] A. Hayakawa, T. Goto, R. Mimoto, Y. Arakawa, T. Kudo, H. Kobayashi, Laminar burning velocity and Markstein length of ammonia/air premixed flames at various pressures, *Fuel* 159 (2015) 98–106.
- [12] A. Valera-Medina, H. Xiao, M. Owen-Jones, W.I.F. David, P.J. Bowen, Ammonia for power, *Prog. Energy Combust. Sci.* 69 (2018) 63–102.
- [13] H. Kobayashi, A. Hayakawa, K.K.A. Somaratne, E.C. Okafor, Science and technology of ammonia combustion, *Proc. Combust. Inst.* 37 (2019) 109–133.
- [14] A. Valera-Medina, F. Amer-Hatem, A.K. Azad, I. Dedoussi, M.D. Joannon, R.X. Fernandes, P. Glarborg, H. Hashemi, X. He, S. Mashurk, J. McGowan, C. Mounaim-Rouselle, A. Ortiz-Prado, J.A. Ortiz-Valera, I. Rossetti, B. Shu, M. Yehia, H. Xiao, M. Costa, A review on ammonia as a potential fuel: from synthesis to economics, *Energy Fuels* 35 (2021) 6964–7029.
- [15] W.H. Moberly, Shock tube study of hydrazine decomposition, *J. Phys. Chem.* 66 (1962) 366–368.
- [16] K.W. Michel, H.G. Wagner, Untersuchung des thermischen Zerfalls und der Oxydation von Hydrazin mit Stoßwellen, *Z. Phys. Chem.* 35 (1962) 392–395.
- [17] R.W. Diesen, Mass spectral studies of kinetics behind shock waves. II. Thermal decomposition of hydrazine, *J. Chem. Phys.* 39 (1963) 2121–2128.
- [18] E.T. McHale, B.E. Knox, H.B. Palmer, Determination of the decomposition kinetics of hydrazine using a single-pulse shock tube, *Symp. (Int.) Combust.* 10 (1965) 341–351.
- [19] K.W. Michel, Thermal decomposition of hydrazine, *Angew. Chem. Internat. Ed.* 4 (1965).
- [20] K.W. Michel, H.G. Wagner, The pyrolysis and oxidation of hydrazine behind shock waves, *Symp. (Int.) Combust.* 10 (1965) 353–364.
- [21] E. Meyer, H.A. Olschewski, J. Troe, H.G. Wagner, Investigation of N_2H_4 and H_2O_2 decomposition in low and high pressure shock waves, *Symp. (Int.) Combust.* 12 (1969) 345–355.
- [22] A.P. Genich, A.A. Zhirnov, G.B. Manelis, Decomposition of hydrazine behind reflected shock waves at high pressures, *Russ. J. Phys. Chem.* 48 (1974) 416–430.
- [23] M. Szwarc, The dissociation energy of the NN bond in hydrazine, *Proc. R. Soc. Lond. A* 198 (1949) 267–284.
- [24] J.A. Kerr, R.C. Sekhar, A.F. Trotman-Dickenson, The pyrolyses of hydrazines and benzylamines. C-C and N-N bond dissociation energies, *J. Chem. Soc.* (1963) 3217–3225.
- [25] I.J. Eberstein, I. Glassman, The gas-phase decomposition of hydrazine and its methyl derivatives, *Symp. (Int.) Combust.* 10 (1965) 365–374.
- [26] P.V. Khe, J.C. Soullignac, R. Lesclaux, Pressure and temperature dependence of amino radical recombination rate constant, *J. Phys. Chem.* 81 (1977) 210–214.
- [27] V.A. Lozovskii, V.A. Nadtochenko, O.M. Sarkisov, S.G. Cheskis, Study of NH_2 radical recombination by intracavity laser spectroscopy, *Kinet. Catal.* 20 (1979) 918–922.
- [28] O.M. Sarkisov, S. Cheskis, V.A. Nadtochenko, E.A. Sviridenkov, V.I. Vedenev, Spectroscopic study of elementary reactions involving oxomethyl, amidogen, and nitrosyl hydride, *Arch. Combust.* 4 (1984) 111–120.

- [29] K. Fagerström, J.T. Jodkowski, A. Lund, E. Ratajczak, Kinetics of the self-reaction and the reaction with OH of the amidogen radical, *Chem. Phys. Lett.* 236 (1995) 103–110.
- [30] M.-K. Bahng, R.G. Macdonald, Determination of the rate constant for the $\text{NH}_2(\text{X}^2\text{B}_1) + \text{NH}_2(\text{X}^2\text{B}_1)$ reaction at low pressure and 293K, *J. Phys. Chem. A* 112 (2008) 13432–13443.
- [31] G. Altinay, R.G. Macdonald, Determination of the rate constants for the $\text{NH}_2(\text{X}^2\text{B}_1) + \text{NH}_2(\text{X}^2\text{B}_1)$ recombination reaction with collision partners He, Ne, Ar, and N_2 at low pressures and 296K. Part 1, *J. Phys. Chem. A* 116 (2012) 1353–1367.
- [32] G. Altinay, R.G. Macdonald, Determination of the rate constants for the $\text{NH}_2(\text{X}^2\text{B}_1) + \text{NH}_2(\text{X}^2\text{B}_1)$ and $\text{NH}_2(\text{X}^2\text{B}_1) + \text{H}$ recombination reactions with collision partners CH_4 , C_2H_6 , CO_2 , CF_4 , and SF_6 at low pressures and 296K. Part 2, *J. Phys. Chem. A* 116 (2012) 2161–2176.
- [33] G. Altinay, R.G. Macdonald, Determination of the rate constants for the $\text{NH}_2(\text{X}^2\text{B}_1) + \text{NH}_2(\text{X}^2\text{B}_1)$ and $\text{NH}_2(\text{X}^2\text{B}_1) + \text{H}$ recombination reactions in N_2 as a function of temperature and pressure, *J. Phys. Chem. A* 119 (2015) 7593–7610.
- [34] P. Glarborg, H. Hashemi, S. Cheski, A.W. Jasper, On the rate constant for $\text{NH}_2 + \text{HO}_2$ and third body collision efficiencies for $\text{NH}_2 + \text{H}(\text{+M})$ and $\text{NH}_2 + \text{NH}_2(\text{+M})$, *J. Phys. Chem. A* 125 (2021) 1505–1516.
- [35] S.J. Klippenstein, L.B. Harding, B. Ruscic, R. Sivaramakrishnan, N.K. Srinivasan, M.-C. Su, J.V. Michael, Thermal decomposition of NH_2OH and subsequent reactions: ab initio transition state theory and reflected shock tube experiments, *J. Phys. Chem. A* 113 (2009) 10241–10259.
- [36] L. Dai, S. Gersen, P. Glarborg, H. Levinsky, A. Mokhov, Experimental and numerical analysis of the autoignition behavior of NH_3 and NH_3/H_2 mixtures at high pressure, *Combust. Flame* 215 (2020) 134–144.
- [37] O.V. Dorofeeva, O.N. Ryzhova, T.A. Suchkova, Enthalpies of formation of hydrazine and its derivatives, *J. Phys. Chem. A* 121 (2017) 5361–5370.
- [38] D. Feller, D.H. Bross, B. Ruscic, Enthalpy of formation of N_2H_4 (hydrazine) revisited, *J. Phys. Chem. A* 121 (2017) 6187–6198.
- [39] C.J. Cobos, A.E. Croce, K. Luther, L. Soelter, E. Tellbach, J. Troe, Experimental and modeling study of the reaction $\text{C}_2\text{F}_4(\text{+M}) \rightarrow \text{CF}_2 + \text{CF}_2(\text{+M})$, *J. Phys. Chem. A* 117 (2013) 11420–11429.
- [40] C.J. Cobos, L. Sölter, E. Tellbach, J. Troe, Falloff curves of the reaction $\text{CF}_3(\text{+M}) \rightarrow \text{CF}_2(\text{+M})$, *J. Phys. Chem. A* 124 (2020) 1235–1239.
- [41] J. Troe, Theory of thermal unimolecular reactions at low pressures: I. Solutions of the master equation, *J. Chem. Phys.* 66 (1977) 4745–4757.
- [42] J. Troe, Theory of thermal unimolecular reactions at low pressures. II. Strong collision rate constants. Applications, *J. Chem. Phys.* 66 (1977) 4758–4775.
- [43] J. Troe, Predictive possibilities of unimolecular rate theory, *J. Phys. Chem.* 83 (1979) 114–126.
- [44] A.I. Maergoiz, E.E. Nikitin, J. Troe, V.G. Ushakov, Classical trajectory and statistical adiabatic channel study of the dynamics of capture and unimolecular bond fission. V. Valence interactions between two linear rotors, *J. Chem. Phys.* 108 (1998) 9987.
- [45] J. Troe, V.G. Ushakov, Revisiting falloff curves of thermal unimolecular reactions, *J. Chem. Phys.* 135 (2011) 054304.
- [46] J. Troe, V.G. Ushakov, Representation of "broad" falloff curves for dissociation and recombination reactions, *Z. Phys. Chem.* 228 (2014) 1–10.
- [47] Y. Zhao, D.G. Truhlar, Comparative DFT study of van der Waals complexes: rare-gas dimers, alkaline-earth dimers, zinc dimer, and zinc-rare-gas dimers, *J. Phys. Chem. A* 110 (2006) 5121–5129.
- [48] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G.A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H.P. Hratchian, A.F. Izmaylov, J. Bloino, G. Zheng, J.L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. Montgomery, J.E. Peralta, F. Ogliaro, M. Bearpark, J.J. Heyd, E. Brothers, K.N. Kudin, V.N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J.C. Burant, S.S. Iyengar, J. Tomasi, M. Cossi, M. Rega, M. Klene, J.E. Knox, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R.E. Gomperts, O. Stratmann, A.J. Yazyev, R. Austin, C. Cammi, J.W. Pomelli, R. Ochterski, R.L. Martin, K. Morokuma, V.G. Zakrzewski, G.A. Voth, P. Salvador, J.J. Dannenberg, S. Dapprich, A.D. Daniels, O. Farkas, J.B. Foresman, J.V. Ortiz, J. Cioslowski, D. J. Fox, Gaussian 09, Gaussian, Wallingford, CT, 2009.
- [49] L.A. Curtiss, P.C. Redfern, K. Raghavachari, Gaussian-4 Theory, *J. Chem. Phys.* 126 (2007) 084108.
- [50] J. Troe, Theory of unimolecular reactions in the fall-off range. I. Strong collision rate constants, *Ber. Bunsenges. Phys. Chem.* 87 (1983) 161–169.
- [51] R.G. Gilbert, K. Luther, J. Troe, Theory of thermal unimolecular reactions in the falloff range. II. Weak collision rate constants, *Ber. Bunsenges. Phys. Chem.* 87 (1983) 169–177.
- [52] R.W. Diesen, W.J. Felmlee, Mass spectral studies of kinetics behind shock waves. I. Thermal dissociation of chlorine, *J. Chem. Phys.* 39 (1963) 2115–2120.
- [53] R.W. Diesen, Mass-spectral studies of kinetics behind shock waves. III. Thermal dissociation of fluorine, *J. Chem. Phys.* 44 (1966) 3662–3666.
- [54] R.W. Diesen, Observation and kinetic investigation of the NF and NF_2 radicals, *J. Chem. Phys.* 41 (1964) 3256–3257.
- [55] D.L. Baulch, J. Duxbury, S.J. Grant, D.C. Montague, Evaluated kinetic data for high temperature reactions. Volume 4. Homogeneous gas phase reactions of halogen- and cyanide-containing species, *J. Phys. Chem. Ref. Data* 10 (1981) Supplement 1.
- [56] R.S. Tranter, B.R. Giri, J.H. Kiefer, Shock tube/time-of-flight mass spectrometer for high temperature kinetic studies, *Rev. Sci. Instrum.* 78 (2007) 034101.
- [57] J.B. Burkholder, S.P. Sander, J. Abbatt, J.R. Barker, C. Cappa, J.D. Crounse, T.S. Dibble, R.E. Huie, C.E. Kolb, M.J. Kurylo, V.L. Orkin, C.J. Percival, D.M. Wilmouth, P.H. Wine, Chemical Kinetics and Photochemical Data for Use in Atmospheric Studies, Evaluation No. 19, NASA/JPL, Pasadena, CA, 2019 Publication 19-5. <http://jpldataeval.jpl.nasa.gov>.
- [58] J.R. Durig, S.F. Bush, E.E. Mercer, Vibrational spectrum of hydrazine-d4 and a Raman study of hydrogen bonding in hydrazine, *J. Chem. Phys.* 44 (1966) 4238–4247.
- [59] T. Kasuya, T. Kojima, Internal motions of hydrazine, *J. Phys. Soc. Jpn.* 18 (1963) 364–368.
- [60] K. Kohata, T. Fukuyama, K. Kuchitsu, Molecular structure of hydrazine as studied by gas electron diffraction, *J. Phys. Chem.* 86 (1982) 602–606.
- [61] J.B. Burkholder, C.J. Howard, A.R.W. McKellar, Fourier transform infrared spectrum of the n2 band of the NH_2 radical, *J. Mol. Spectrosc.* 127 (1988) 415–424.
- [62] A.R.W. McKellar, M. Vervloet, J.B. Burkholder, C.J. Howard, A combined analysis of the n1, n3, and 2n2 vibrational states of the NH_2 radical using Fourier transform absorption and emission data, *J. Mol. Spectrosc.* 142 (1990) 319–335.
- [63] P.B. Davies, D.K. Russell, B.A. Thrush, H.E. Radford, Rotational and hyperfine parameters of $\text{NH}_2(\text{X}^2\text{B}_1)$ from LMR spectra, *Chem. Phys. Lett.* 42 (1976) 35–38.
- [64] E. Wilhelm, R. Battino, Estimation of Lennard-Jones (6,12) pair potential parameters from gas solubility data, *J. Chem. Phys.* 55 (1971) 4012–4017.
- [65] H. Hippler, J. Troe, H.J. Wendelken, Collisional deactivation of vibrationally highly excited polyatomic molecules. II. Direct observations for excited toluene, *J. Chem. Phys.* 78 (1983) 6709–6717.
- [66] C.J. Cobos, J. Troe, The dissociation-recombination system $\text{CH}_4 + \text{M} \rightleftharpoons \text{CH}_3 + \text{H} + \text{M}$: II. Evaluation of experiments up to 5000K and temperature dependence of $\langle \Delta E \rangle$, *Z. Phys. Chem* 176 (1992) 161–171.
- [67] P. Glarborg, H. Hashemi, P. Marshall, Challenges in Kinetic Modeling of Ammonia Pyrolysis, *Fuel Commun* 10 (2022) 100049.
- [68] A. Grinberg Dana, K.B. Moore III, A.W. Jasper, W.H. Green, Large intermediates in hydrazine decomposition: a theoretical study of the N_3H_5 and N_4H_6 potential energy surfaces, *J. Phys. Chem. A* 123 (2019) 4679–4692.
- [69] P. Diévert, L. Catoire, Contributions of experimental data obtained in concentrated mixtures to kinetic studies: application to mono methyl hydrazine pyrolysis, *J. Phys. Chem. A* 124 (2020) 6214–6236.
- [70] Y. Gao, I.M. Alecu, H. Hashemi, P. Glarborg, P. Marshall, Reactions of hydrazine with the amidogen radical and atomic hydrogen, *Proc. Combust. Inst.* 39, in press.2022
- [71] N. Kanno, T. Kito, Theoretical study on the hydrogen abstraction reactions from hydrazine derivatives by H atom, *Int. J. Chem. Kinet.* 52 (2020) 548–555.
- [72] P. Marshall, G. Rawling, P. Glarborg, New reactions of diazene and related species for modelling combustion of amine fuels, *Mol. Phys.* 119 (2021) e1979674.
- [73] E. Meyer, H.G. Wagner, Zum Mechanismus des thermischen Hydrazinzerfalls, *Z. Phys. Chem. NF* 89 (1974) 329–331.